Taiwanese Bankruptcy Prediction

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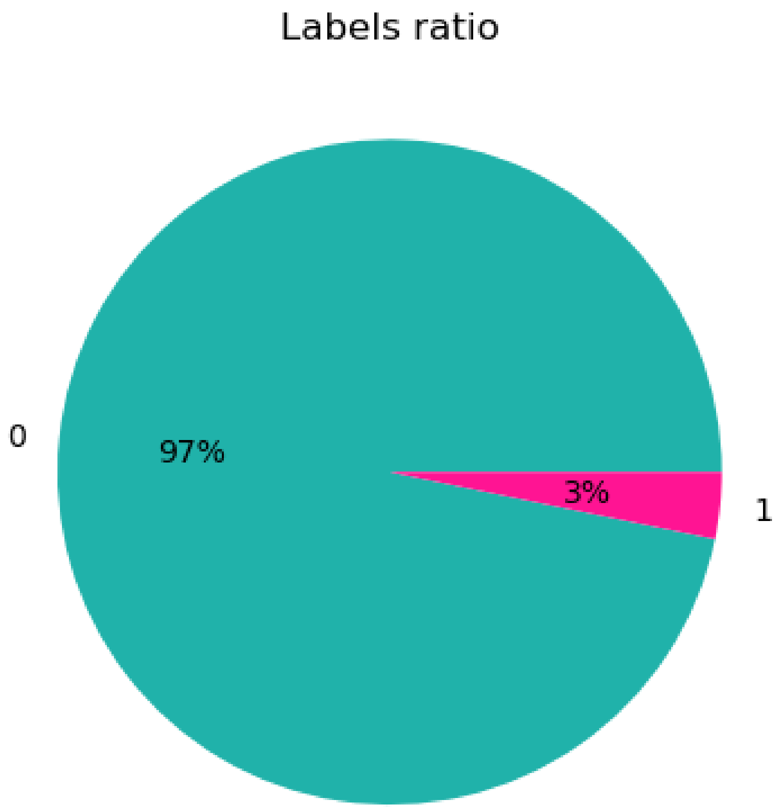
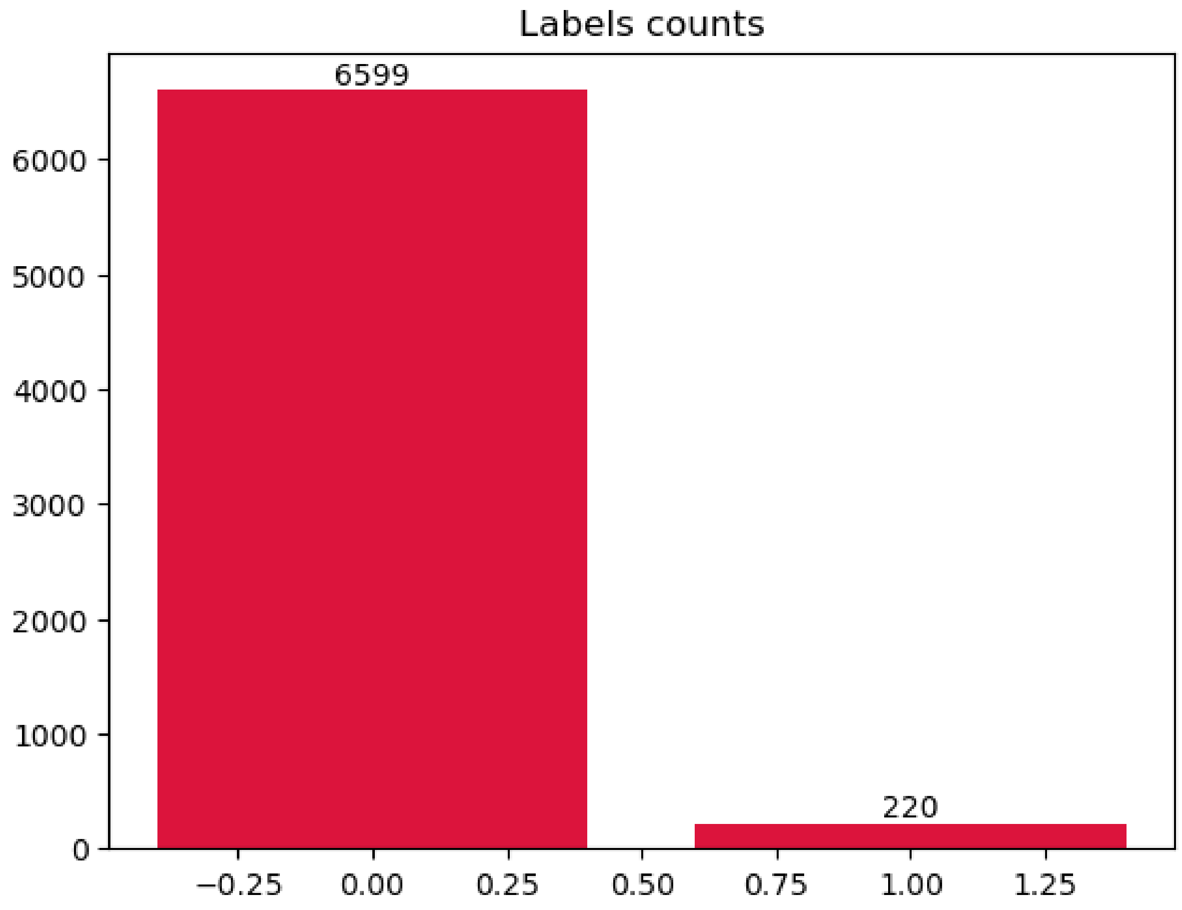
Data characteristic

- 6819 companies
- 95 features
- Target: **Bankrupt?** {0, 1}
- No missing values
- Numerical data / some categorical features like "net income flag"

Main problem: imbalance of predicted variable





Strategy

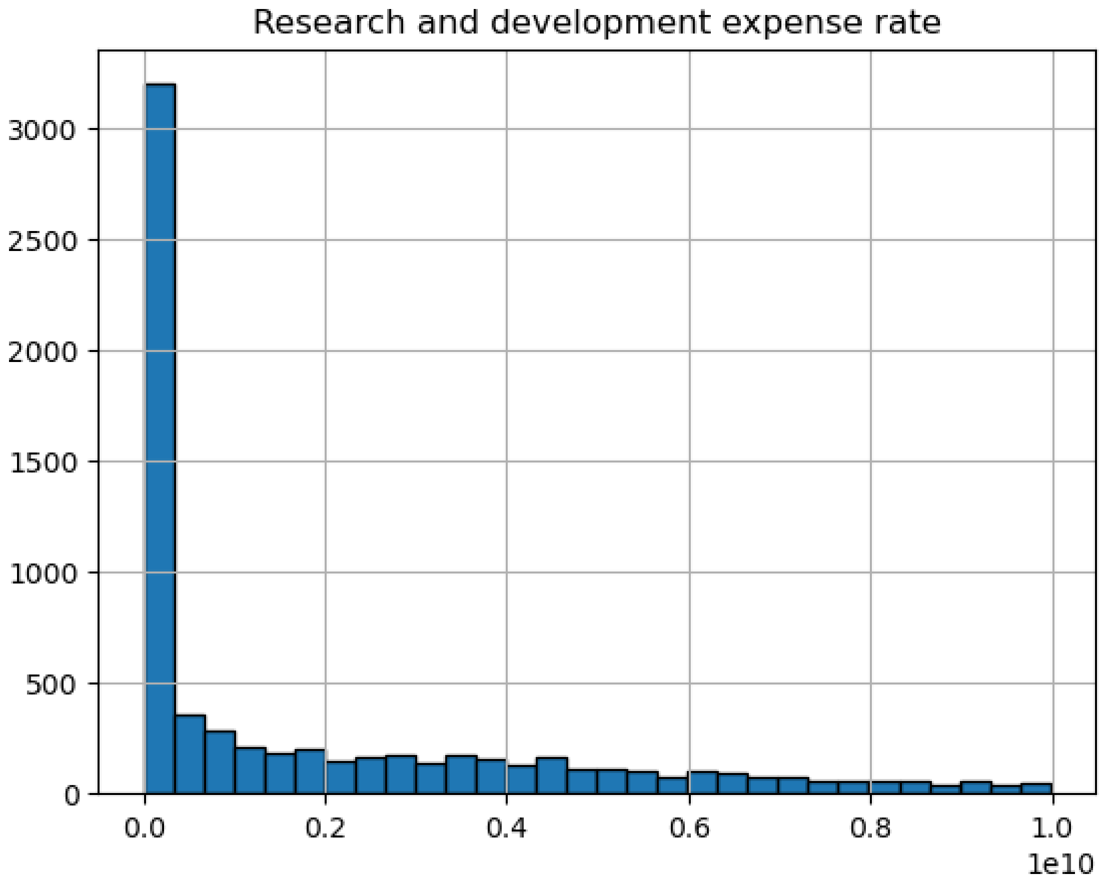
- Datasets split:
  - Train: 70%
  - Validation: 15%
  - Test: 15%

- Handling imbalance:
  - Using weights for classes {0, 1}
  - Using different sampling strategies:
    - Synthetic minority oversampling (SMOTE)
    - TomekLink
    - SMOTEK

Metrics

- Precision
- Recall
- F1-Score
- Maro AVG F1-Score
- Average Precision Score

- Experimenting with different feature engineering techniques and data processing



Undersampling bankruptcy prediction: Taiwan bankruptcy data

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Abstract

Machine learning models have increasingly been used in bankruptcy prediction. However, the observed historical data of bankrupt companies are often affected by data imbalance, which causes incorrect prediction, resulting in substantial economic losses. Many studies have proposed the insolvency imbalance problem, but little attention has been paid to the effect of the undersampling technology. Therefore, a framework is used to spot-check algorithms quickly and combine which undersampling method and classification model performs best. The results show that Naive Bayes (NB) after Edited Nearest Neighbors (ENN) has the best performance, with an F2-measure of 0.423. In addition, by changing the undersampling rate of the cluster centroid-based method, we find that the performance of the Linear Discriminant Analysis (LDA) and Naive Bayes (NB) are affected by the undersampling rate. Neither of them is uniformly declining, and LDA has higher performance when the undersampling rate is 30%. This study accordingly provides another perspective and a guide for future design.

Figures

Altman Z-score

$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$.

X_1 = ratio of working capital to total assets. Measures liquid assets in relation to the size of the company.

X_2 = ratio of retained earnings to total assets. Measures profitability that reflects the company's age and earning power.

X_3 = ratio of earnings before interest and taxes to total assets. Measures operating efficiency apart from tax and leveraging factors. It recognizes operating earnings as being important to long-term viability.

X_4 = ratio of market value of equity to book value of total liabilities. Adds market dimension that can show up security price fluctuation as a possible red flag.

X_5 = ratio of sales to total assets. Standard measure for total asset turnover (varies greatly from industry to industry).

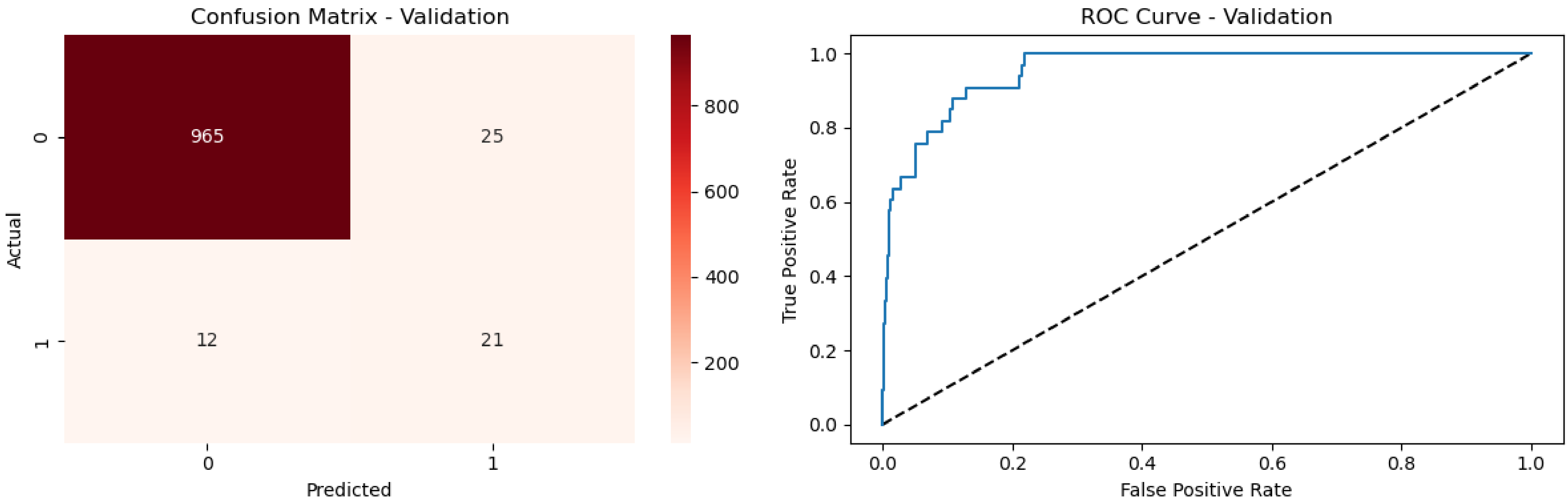
```
z_score += 1.2 * df_eng['Working Capital to Total Assets']
z_score += 1.4 * df_eng['Retained Earnings to Total Assets']
z_score += 3.3 * df_eng['Operating Profit Rate'] * df_eng['Total Asset Turnover']
z_score += 0.6 * df_eng['Total assets to GNP price']
z_score += 1.0 * df_eng['Total Asset Turnover']
```

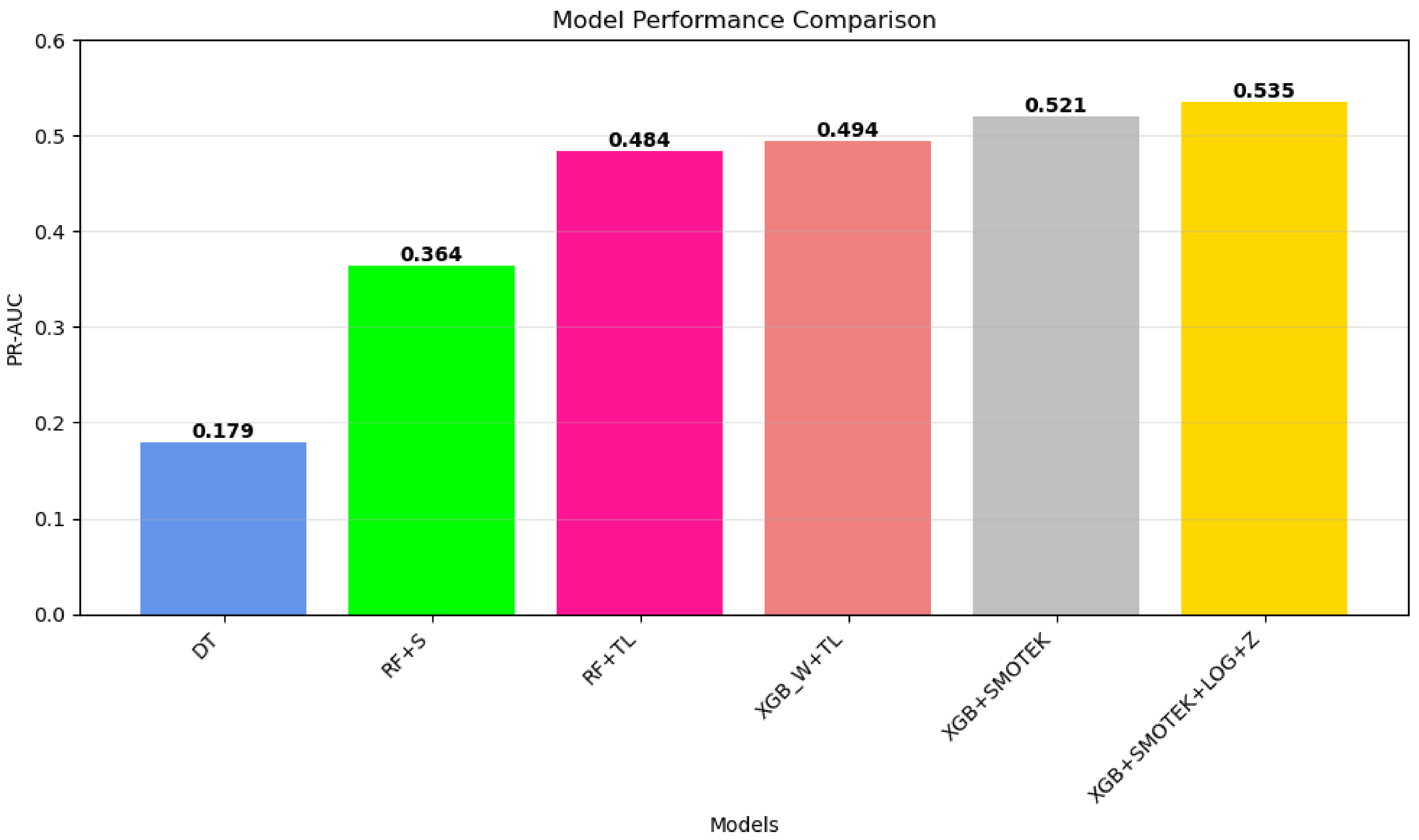
Models

- Logistic Regression
- Logistic Regression + weights
- Decision Tree
- Decision Tree + weights
- Decision Tree + TomekLink
- Random Forests + TomekLink
- XGBoost + TomekLink
- XGBoost + SMOTEK
- Random Forests + SMOTEK
- XGBoost + SMOTEK + Altman Z + Log Transform

MODEL	PRECISION		RECALL		F1-SCORE		MACRO AVG F1 - score	AVRAGE PRECISION SCORE
	0	1	0	1	0	1		
Logistic regression	0.96	0.25	1.00	0.03	0.98	0.05	0.51	0.06
Logistic regression + weights	0.97	0.07	0.78	0.44	0.86	0.12	0.49	0.08
Decision Tree	0.98	0.41	0.98	0.38	0.98	0.39	0.69	0.179
Decision Tree + TomekLink	0.97	0.36	0.98	0.31	0.98	0.33	0.65	0.138
XGBoost + weights + TomekLink	0.97	**0.59**	0.99	0.33	0.98	0.43	**0.70**	**0.494**
Random Forest + Tomek Link	0.97	**0.75**	1.00	0.15	0.98	0.26	0.62	0.484
XGBoost + SMOTEK	0.98	0.56	0.99	0.38	0.98	0.45	**0.72**	**0.521**
Random Forest + SMOTEK	0.99	0.32	0.95	**0.64**	0.96	0.42	0.69	0.364
XGBoost + SMOTEK + Altman Z + log transform on skewed	0.98	0.40	0.97	**0.52**	0.98	0.45	**0.71**	**0.5353**

XGBoost + SMOTEK + Altman Z + log transform on skewed distributions




Model Performance Comparison
PR-AUC
Models
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.179
0.364
0.484
0.494
0.521
0.535
DT
RF+S
RF+TL
XGB_W+TL
XGB+SMOTEK
XGB+SMOTEK+LOG+Z